

CogniPDF AI

Enterprise Multi-Document RAG & Neural Voice Assistant — Master Technical Documentation

Version: 2.4.0 | Framework: Flask / LangChain / Groq | Date: August 2026

1. Executive Summary

CogniPDF AI is a production-ready, multi-document **Retrieval-Augmented Generation (RAG)** application. It provides high-performance document ingestion, local vectorization, and sub-second natural language inference across heterogeneous PDF files (including native digital PDFs, scanned documents, contracts, scientific papers, and financial spreadsheets). The system incorporates local OCR fallbacks, local dense vector embeddings, sub-second LPU reasoning via Groq Cloud, and dual-engine neural speech synthesis.

2. System Architecture & Ingestion Pipeline

The architecture is organized into four decoupled processing layers:

- **1. Multi-Tier Ingestion & OCR:** Uploaded PDFs are parsed via *PyMuPDF (fitz)* for sub-millisecond text extraction. If a scanned image-only PDF is detected (zero text layer), the engine automatically triggers local *RapidOCR (ONNX Runtime)* to perform image character recognition.
- **2. Semantic Chunking:** Cleaned textual passages are partitioned using LangChain's *RecursiveCharacterTextSplitter* (chunk size = 1,000 characters, overlap = 200 characters).
- **3. Local Embedding & Vector Store:** Dense 384-dimensional vector representations are generated via *all-MiniLM-L6-v2* running on local ONNX Runtime (zero cloud embedding costs), and indexed into in-memory *ChromaDB* with metadata (filename, page number, chunk ID).
- **4. Retrieval & LLM Synthesis:** User queries undergo cosine similarity search. Top-K relevant contexts are injected into a structured system prompt dispatched to *Groq LPUs* for sub-second synthesis with exact source citations and 3 contextual follow-up questions.
- **5. Neural Speech Synthesis:** AI responses can be synthesized into studio-quality audio streams using Microsoft Neural Voices (Edge-TTS) or ElevenLabs.

3. AI Models & Purpose Reference

Model Name	Category	Provider	Role & Purpose
llama-3.3-70b-versatile	LLM (Reasoning)	Meta / Groq	Primary reasoning engine. Synthesizes answers, quotes exact passages, and generates citations.
groq/compound-mini	LLM (Fast)	Groq	Ultra-low latency reasoning model for quick summaries and quick prompt chips.
openai/gpt-oss-120b	LLM (Flagship)	Groq	Flagship open-weight model for complex document comparisons and multi-document reasoning.
qwen/qwen3.6-27b	LLM (Math/Multi)	Qwen / Groq	Multilingual & mathematical reasoning across technical formulas and foreign language texts.
all-MiniLM-L6-v2	Embedding (384-dim)	ONNX / Local	Local semantic vectorization model. Generates dense embeddings offline with zero API cost.
nomic-embed-text-v1.5	Embedding (768-dim)	Nomic AI	Optional cloud vector embedding with 8,192-token context window for long documents.
RapidOCR (PP-OCRv4)	OCR Engine	ONNX Runtime	Extracts text from scanned/image-only PDFs locally without external cloud API charges.
en-US-AriaNeural	Neural TTS	Microsoft	Default studio-quality neural voice for hands-free audio playback.
ElevenLabs Multilingual v2	Voice Cloning	ElevenLabs	Optional hyper-realistic voice personalities for custom narration.

4. Complete Technology Stack

- **Backend Framework:** Flask 3.1.3 (REST API), Werkzeug 3.1.8
- **Production WSGI Server:** Gunicorn 26.2.0 (Configured with --workers 1 --threads 4 --timeout 120 for synchronized thread-safe state)
- **RAG Pipeline:** LangChain 0.1+, LangChain Groq, ChromaDB 1.5+ (In-Memory Vector DB)
- **Concurrency Control:** Python threading.Lock() mutex protecting parallel upload/delete transactions
- **PDF Extraction:** PyMuPDF 1.28+ (C-accelerated), pdfplumber 0.11+, pypdf 6.16+
- **Local OCR Engine:** RapidOCR 1.4+ (ONNX Runtime, PP-OCRv4 model)
- **Speech Synthesis:** edge-tts 7.2+, ElevenLabs REST API, Web Speech API (Browser Voice Dictation)
- **Frontend:** Vanilla HTML5 / ES6+ JavaScript / Modern CSS Design Tokens (Segmented Mobile Tab Navigation)
- **Markdown & Code Highlighting:** Marked.js with Highlight.js

5. REST API Endpoints Reference

Endpoint	Method	Payload	Description
/status	GET	None	Returns system health, document list, page/chunk stats, and Groq status.
/upload	POST	multipart/form-data	Ingests, parses, chunks, embeds, and indexes a PDF document into ChromaDB.
/ask	POST	JSON {question, model}	Performs Top-K search and returns answer with citations, latency, and follow-ups.
/tts	POST	JSON {text, voice}	Generates and streams high-fidelity MP3 neural audio.
/delete-file	POST	JSON {filename}	Removes document and purges its vector embeddings from ChromaDB.
/clear	POST	None	Purges all documents, resets vector store, and clears chat history.
/clear-history	POST	None	Resets conversational memory without deleting indexed documents.
/update-config	POST	JSON {api_key, model}	Dynamically updates runtime Groq API Key and default model.

6. Setup & Deployment Guide

Environment Variables (.env):

GROQ_API_KEY = gsk_your_key_here | GROQ_MODEL = llama-3.3-70b-versatile | ELEVENLABS_API_KEY = optional_key

Local Execution:

1. pip install -r requirements.txt
2. python app.py (Server runs at http://127.0.0.1:5000)

Render Cloud Production:

- Start Command: gunicorn app:app --workers 1 --threads 4 --timeout 120
- Build Command: pip install -r requirements.txt